

6COSC022W Coursework 2 (2024/25)

Module leader	Indrajith Ekanayake
Unit	Coursework 2
Weighting:	60%
Qualifying mark	30%
Description	Design of web application and essay
Learning Outcomes Covered in Assignment:	L02. LO3, this LO4
Handed Out:	
Due Date	30th April 2026 , 11.59pm IST
Expected deliverables	Electronic document and web application deployed on university servers
Method of Submission:	online via Blackboard
Type of Feedback and Due Date:	Feedback will be given during viva following deadline submission and via Blackboard rubric. All marks will remain provisional until formally agreed by an Assessment Board.

Assessment regulations

Refer to section 4 of the “How you study” guide for undergraduate students for a clarification of how you are assessed, penalties and late submissions, what constitutes plagiarism etc.

Penalty for Late Submission

If you submit your coursework late but within 24 hours or one working day of the specified deadline, 10 marks will be deducted from the final mark, as a penalty for late submission, except for work which obtains a mark in the range 40 – 49%, in which case the mark will be capped at the pass mark (40%). If you submit your coursework more than 24 hours or more than one working day after the specified deadline you will be given a mark of zero for the work in question unless a claim of Mitigating Circumstances has been submitted and accepted as valid.

It is recognized that on occasion, illness or a personal crisis can mean that you fail to submit a piece of work on time. In such cases you must inform the Campus Office in writing on a mitigating circumstances form, giving the reason for your late or non-submission. You must provide relevant documentary evidence with the form. This information will be reported to the relevant Assessment Board that will decide whether the mark of zero shall stand. For more detailed information regarding University

Assessment Regulations, please refer to the following website:
<http://www.westminster.ac.uk/study/current-students/resources/academic-regulations>

Coursework Description

Alumni Influencers - Web Client

Background Scenario

Part 2: University Analytics & Intelligence Dashboard

Weighting: 60%

Prerequisites: Successful completion of Part 1

Background & Business Context

Universities traditionally struggle with understanding what happens to graduates after they leave. Current methods rely on:

- Annual graduate surveys with 30-40% response rates
- Self-reported data collected 6-12 months after graduation
- Historical information that's already outdated when analysed
- Limited insight into career progression beyond first destination
- No visibility into post-graduation professional development

This creates a critical gap: by the time universities identify skills gaps or emerging industry requirements, they're already 2-3 years behind. Graduates discover these gaps themselves and independently acquire certifications, attend bootcamps, and complete professional courses to remain competitive.

The Alumni Influencers Solution

Part 1 created a platform where alumni maintain detailed, current professional profiles to compete for visibility. This generates a living, real-time dataset of actual graduate outcomes.

Part 2 builds the University Analytics Dashboard - a dedicated client application that transforms this alumni data into actionable intelligence for curriculum development and strategic planning.

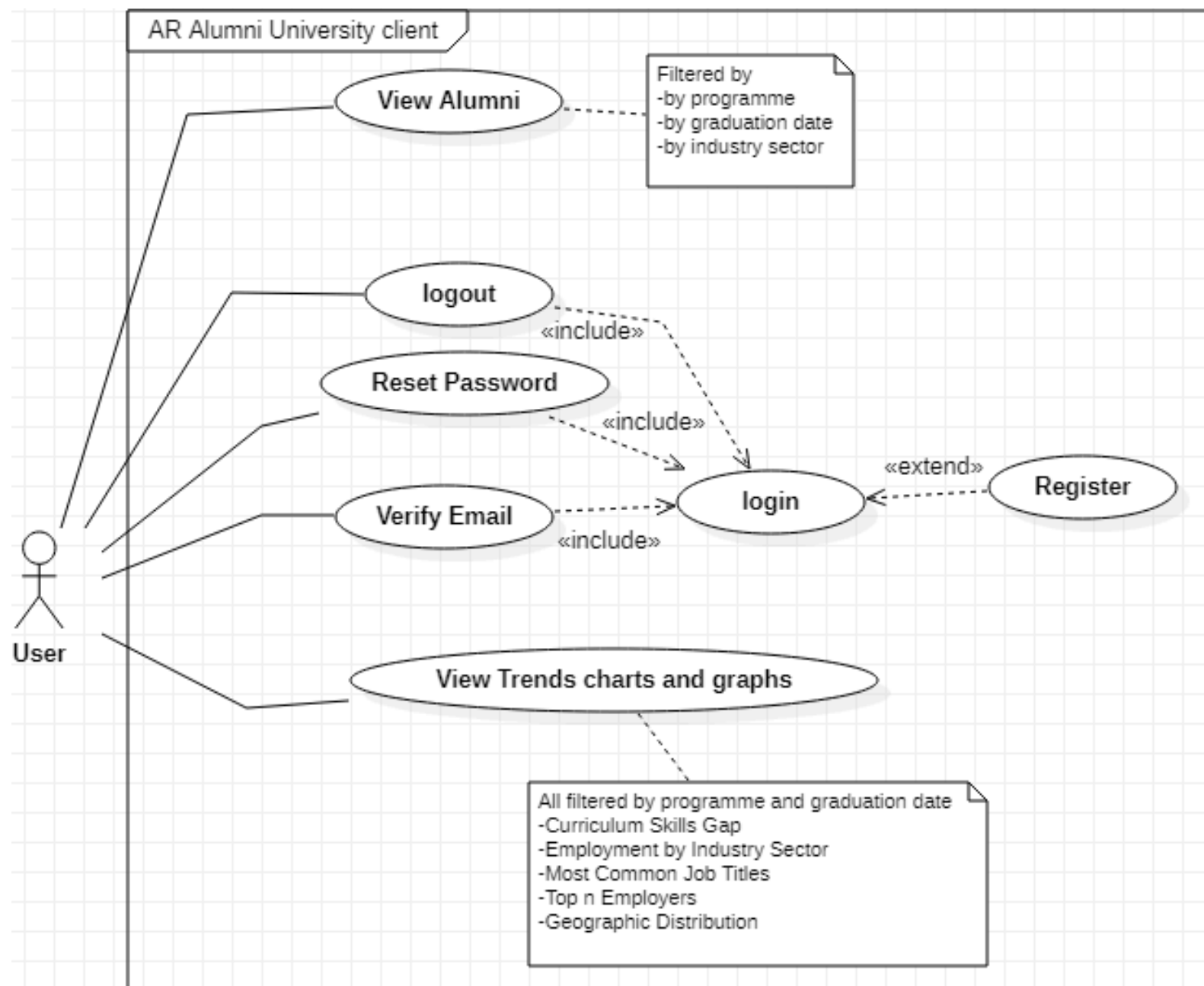
Real-World Intelligence Examples:

Scenario 1: Skills Gap Detection The University Computer Science programme doesn't include Docker/Kubernetes in the curriculum. Within 18 months of graduation, 73% of CS alumni have independently acquired container orchestration certifications. This signals a critical curriculum gap that the university can address before the next cohort graduates unprepared.

Scenario 2: Emerging Career Pathways Data shows 23% of Business Management graduates now work in "Data Analytics" roles despite no formal data science training. Alumni have independently completed Python, SQL, and Tableau courses. This reveals an emerging career pathway the university isn't preparing students for.

Scenario 3: Industry Demand Tracking Over the past 6 months, cloud certifications (AWS, Azure, GCP) have increased 156% among graduates. The university can proactively integrate cloud computing earlier in the curriculum rather than reactively adding it years later.

Scenario 4: Professional Development Trends The most completed post-graduation courses are Agile/Scrum certifications (31% of graduates), suggesting universities should integrate agile methodologies into project modules rather than leaving graduates to learn these critical workplace practices independently.



Part 2 Features Overview

Core Features You Will Build

1. **Registration & Authentication**
 - Email-based registration (university domain required)
 - Email verification system
 - Secure login/logout with sessions
 - Password reset functionality
2. **Complete Webpages**
 - Dashboard
 - Graphs Pages
 - View Alumni by programme, graduation date and industry sector.
3. **Blind Bidding System**
 - Place bids without seeing highest bid amount
 - Receive win/lose status feedback
 - Update bids (increase only)
 - Monthly limit enforcement (3 features per month)
 - Automated midnight winner selection (Mark the profile that will be displayed)
4. **Security to restrict client Access through bearer tokens**
 - Decide how different clients will access the data
 - View usage statistics i.e. number of times and time stamps of clients logging in and usage of keys / tokens and endpoints accessed
 - **API key Scoping:** API keys should be scoped to specific client applications. For example, the university analytics dashboard should only have permissions like ["read:alumni", "read:analytics"], while the mobile AR app should have its own separate key with permissions like ["read:alumni_of_day"]. This ensures that a compromised analytics dashboard key cannot be used to access mobile AR endpoints, and vice versa.

Client Platform	API Key Permissions	Cannot Access
Analytics Dashboard	read:alumni read:analytics	AR app endpoints
Mobile AR App	read:alumni_of_day	Analytics endpoints

Coursework 2 Marking Scheme

Dashboard Layout & Navigation (5 marks)

- **Excellent (5):** Professional dashboard with intuitive navigation, responsive design, clear section organization, proper loading states, user-friendly interface
- **Good (3-4):** Functional dashboard with minor UX issues
- **Fair (1-2):** Basic dashboard with navigation problems
- **Poor (0):** Confusing or non-functional interface

Data Visualization & Charts (30 marks)

- **Excellent (24-30):** Multiple chart types (bar, line, pie, radar, doughnut) displaying alumni data from database via API, interactive charts with tooltips and legends, proper axis labelling, color-coded insights (critical/significant/emerging gaps), responsive charts, smooth animations, at least 6-8 different charts showing various metrics
- **Good (16-23):** Working charts with minor visualization issues, most chart types implemented, data correctly fetched from API
- **Fair (8-15):** Basic charts with limited interactivity, poor readability, missing chart types, or hardcoded data
- **Poor (0-7):** Missing or non-functional visualizations, charts not displaying data from API

Export & Report Generation (5 marks)

- **Excellent (5):** Export filtered data to CSV/PDF, generate custom reports with selected metrics, save filter presets, downloadable chart images
- **Good (3-4):** Basic export functionality with minor limitations
- **Fair (1-2):** Limited export capabilities
- **Poor (0):** Missing export capabilities

API Key Scoping & Permissions (5 marks)

- **Excellent (5):** Granular permission system (read:alumni, read:analytics, read:donations, read:alumni_of_day), permissions enforced on all endpoints, different keys for different clients (Analytics Dashboard vs. AR App), proper 403 responses for unauthorized access, clear documentation of permission levels
- **Good (3-4):** Working permission system with minor enforcement gaps
- **Fair (1-2):** Basic permissions with insufficient granularity
- **Poor (0):** No permission scoping or enforcement

Security Implementation (10 marks)

Password Security (2.5 marks)

- **Excellent (2.5):** Bcrypt hashing with appropriate salt rounds (10+), password strength validation (min length, complexity), secure password storage
- **Good (1.5-2):** Password hashing implemented with minor security gaps
- **Fair (0.5-1):** Basic password hashing with security concerns
- **Poor (0):** Passwords stored in plain text or weak hashing

Input Validation & Sanitization (2.5 marks)

- **Excellent (2.5):** Comprehensive input validation on all forms, SQL/NoSQL injection prevention, XSS protection, data sanitization
- **Good (1.5-2):** Good input validation with minor gaps
- **Fair (0.5-1):** Basic validation with security vulnerabilities
- **Poor (0):** No input validation or sanitization

Authentication Token Security (2.5 marks)

- **Excellent (2.5):** Secure token generation (verification, password reset, API keys), cryptographically random, proper expiry, single-use tokens where appropriate
- **Good (1.5-2):** Token security implemented with minor issues
- **Fair (0.5-1):** Basic token implementation with security gaps
- **Poor (0):** Insecure or predictable tokens

Security Headers & Protection (2.5 marks)

- **Excellent (2.5):** Helmet.js or equivalent implemented, CORS properly configured, CSRF protection, rate limiting on sensitive endpoints
- **Good (1.5-2):** Most security headers and protections in place
- **Fair (0.5-1):** Basic security measures with gaps
- **Poor (0):** No security headers or protections

Data Management (7 marks)

Excellent (6-7 marks)

- Database in 3NF with proper normalization
- Appropriate security fields (password hashing, tokens, API keys)
- API key storage with permissions and usage tracking
- Well-structured relationships between tables/collections
- Efficient queries and appropriate indexing
- Proper foreign key constraints and data integrity

Good (4-5 marks)

- Good database structure with minor normalization issues
- Most security fields implemented correctly
- Basic API key tracking present
- Adequate relationships between tables/collections

Fair (2-3 marks)

- Basic database structure with normalization gaps
- Some security fields missing or poorly implemented
- Limited API key tracking functionality
- Weak relationships between tables/collections

Poor (0-1 marks)

- Poor database design
- Missing security fields
- No API key tracking
- Minimum required tables/collections only

Architecture & Documentation (8 marks)

Excellent (7-8 marks)

- Deliberate architecture design with clear component separation (frontend/backend/database)
- Complete .env.example file with all required environment variables documented
- Detailed code comments explaining complex logic and algorithms
- Database schema documentation with entity relationships

- API endpoint documentation with request/response examples
- Clear README with setup instructions

Good (5-6 marks)

- Some architecture documentation present
- Environment variables listed but lacking detailed explanations
- Code comments present for most complex sections
- Basic API documentation

Fair (2-4 marks)

- Missing or unclear architecture documentation
- Environment variables not properly documented
- Sparse code comments
- Limited or no API documentation

Poor (0-1 marks)

- No architecture documentation or diagrams
- Environment variables not documented
- Missing code comments
- No API documentation

Viva Presentation (30 marks)

Application Demonstration (10 marks)

- Excellent (9-10): Clear demonstration of all features, smooth execution, well-organized presentation
- Good (7-8): Good demonstration of most features with minor issues
- Fair (5-6): Basic demonstration with some technical problems
- Poor (0-4): Incomplete or unclear demonstration

Code Explanation (10 marks)

- Excellent (9-10): Clear explanation of code structure, design patterns, and implementation details
- Good (7-8): Good explanation of main code components
- Fair (5-6): Basic code explanation with some gaps
- Poor (0-4): Poor or incomplete code explanation

Security & Design Justification (10 marks)

- Excellent (9-10): Comprehensive explanation of security implementation choices, architectural decisions, performance considerations, and testing approach
- Good (7-8): Good explanation of most major decisions
- Fair (5-6): Basic explanation of some decisions
- Poor (0-4): Poor or missing justification

Deliverables:

1. Github Repo. It is the student's responsibility to ensure that the repo is accessible to the marker.
2. Viva (Failure to show up to defend your work will result in a grade of Zero.)

Use of Generative AI is forbidden in this coursework.

If you need any clarification about any of these issues, please contact me by email (at indrajith.e@iit.ac.lk) or in the tutorials or make an appointment to see me in my office.

END DOCUMENT